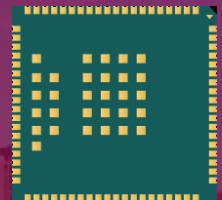


# Quectel EC600M Series

IoT/ M2M-optimized

LTE Cat 1 bis Module



## EC600M-CN-LE\_QuecPython Release Notes

### LTE Standard Module Series

Rev. EC600M-CN-LE\_QuecPython\_Firmware\_Release\_Notes\_V0604

Date: 2024-11-11

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## 1. Release Content

This document provides the Release Notes for EC600M-CN-LE\_QuecPython. The current release includes the following firmware packages.

| Package   | Firmware Version             |
|-----------|------------------------------|
| Firmware  | EC600MCNLER06A04M08_OCPU_QPY |
| Toolchain | Helios_toolchain_1.4.1.exe   |

## 2. Matters Needing Attention

| SN  | Item                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [1] | To prevent the frequently erase and write of SIM card due to the frequent changes in network information, affecting the service life of the SIM card, the network information is not updated by default, and GUTI and other network information are saved when the module is turned off normally with <i>Power.powerDown()</i> . The normal shutdown method is recommended. |
| [2] | To extend the service life of flash, it is recommended that the total number of operations related to power-on/off, CFUN switching, SIM card hot plugging, or dual-SIM switching should not exceed 30 times per day.                                                                                                                                                        |
| [3] | Please make sure that the APN has been set correctly before network registration.                                                                                                                                                                                                                                                                                           |
| [4] | There are two upgrade packages for upgrading via mini-system.                                                                                                                                                                                                                                                                                                               |
| [5] | For UART transmission, it is recommended to add hardware flow control to reduce the risk of data loss during transmission.                                                                                                                                                                                                                                                  |
| [6] | The module turns on the remap function by default. This function is suitable for most operators' LTE network scenarios.                                                                                                                                                                                                                                                     |
| [7] | During the FOTA upgrade process, it is necessary to ensure the stable power supply of the module. If the power is disconnected during the upgrade, there is a small probability that the flash will be damaged.                                                                                                                                                             |
| [8] | During the module shut-down process, the main UART output garbled data when it is turned off.                                                                                                                                                                                                                                                                               |
| [9] | The 32 KHz (clock source) output from the module has an error within 20 PPM. Do not provide it to the peripherals with high accuracy demand.                                                                                                                                                                                                                                |

## 3. Release History

### 3.1. Firmware Release History

| Firmware Version             | Description     |
|------------------------------|-----------------|
| EC600MCNLER06A04M08_OCPU_QPY | Mass production |
| EC600MCNLER06A03M08_OCPU_QPY | Mass production |
| EC600MCNLER06A02M08_OCPU_QPY | Mass production |

### 3.2. New Features

| EC600MCNLER06A04M08_OCPU_QPY |                                                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Item                         | Brief Description                                                                                                                                                                                                             |
| FOTA                         | Add <i>ipvtype</i> to the constructors of the request and app_fota modules. The default value of <i>ipvtype</i> is 0, that is, IPv4, and <i>ipvtype</i> can be set to 1, that is, IPv6.                                       |
| HTTP                         | Support IPv6 in the request module.                                                                                                                                                                                           |
| I2C                          | Add the I2C3 interface, with SLC connected to Pin57 and SDA connected to Pin58. The existing I2C1 interface has SLC connected to Pin75 and SDA to Pin74.                                                                      |
| SSL                          | Add the feature of CA certificate configuration in the ussl module.                                                                                                                                                           |
| General                      | Add <i>utime.localtime_ex()</i> and <i>utime.mktime_ex()</i> to solve the problem that the old functions may cause time conversion errors in scenarios of non-UTC+8 zones because they are only applicable to the UTC+8 zone. |
| General                      | Add the 32 KHz clock signal output feature.                                                                                                                                                                                   |
| EC600MCNLER06A03M08_OCPU_QPY |                                                                                                                                                                                                                               |
| Item                         | Brief Description                                                                                                                                                                                                             |
| Network                      | Add <i>nic.status()</i> to query the status of the W5500 Ethernet NIC.                                                                                                                                                        |
| General                      | Add the half-hour timekeeping functions <i>utime.setTimeZoneEx()</i> and <i>utime.getTimeZoneEx()</i> .                                                                                                                       |
| EC600MCNLER06A02M08_OCPU_QPY |                                                                                                                                                                                                                               |
| Item                         | Brief Description                                                                                                                                                                                                             |
| /                            | /                                                                                                                                                                                                                             |

### 3.3. Improved Features

#### EC600MCNLER06A04M08\_OCPU\_QPY

| Item      | Brief Description                                                                            |
|-----------|----------------------------------------------------------------------------------------------|
| Low Power | Add <i>misc.Power.camVDD2V8Enable()</i> to set the voltage output from the cam_vdd interface |
| PING      | Limit the range of <i>SIZE</i> for the PING feature.                                         |

#### EC600MCNLER06A03M08\_OCPU\_QPY

| Item     | Brief Description                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MiniFOTA | Solve the problem of firmware download failure caused by network registration using blank APN in specific areas through setting the user-configured APN information before the MiniFOTA network is established. |
| Network  | Solve the problem of USBnet function failure.                                                                                                                                                                   |
| (U)SIM   | Solve the problem that the operator name could not be returned after the execution of <i>net.operatorName()</i> or <b>AT+COPS?</b> when a SIM card of China Broadcast Network Co., Ltd was inserted.            |
| General  | Solve the problem that one extra row and one extra column were configured when the matrix keyboard was initialized.                                                                                             |

#### EC600MCNLER06A02M08\_OCPU\_QPY

| Item | Brief Description |
|------|-------------------|
| /    | /                 |

### 3.4. Known Issues

| Item | Brief Description |
|------|-------------------|
| /    | /                 |

#### NOTE

Verification Environment is shown below. For more details, please contact Quectel technical support.  
 USB Driver: Quectel\_Windows\_USB\_DriverA\_Customer\_V1.1.13  
 QPYcom Tool: QPYcom\_V3.6

## 4. Functions List

| Category          | Item           | Supported Version (Since)    | Note |
|-------------------|----------------|------------------------------|------|
| Basic Function    | (U)SIM         | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Data Call      | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | FILE           | EC600MCNLER06A02M08_OCPU_QPY | /    |
| Protocol Function | ALiYun         | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | HTTP           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | HTTPS          | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | MQTT           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | NITZ           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | NTP            | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | PING           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | SSL            | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | TCP/UDP        | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Tencent Cloud  | EC600MCNLER06A02M08_OCPU_QPY | /    |
| Special Function  | ADC            | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Alarm          | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Audio Playback | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Audio Record   | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Camera         | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | Device Info    | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | EINT           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | FOTA           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | GNSS           | EC600MCNLER06A02M08_OCPU_QPY | /    |
|                   | GPIO           | EC600MCNLER06A02M08_OCPU_QPY | /    |

|                           |                              |   |
|---------------------------|------------------------------|---|
| I2C                       | EC600MCNLER06A02M08_OCPU_QPY | / |
| Low Power                 | EC600MCNLER06A02M08_OCPU_QPY | / |
| MiniFOTA                  | EC600MCNLER06A02M08_OCPU_QPY | / |
| Modem                     | EC600MCNLER06A02M08_OCPU_QPY | / |
| PWM                       | EC600MCNLER06A02M08_OCPU_QPY | / |
| SIM Detection             | EC600MCNLER06A02M08_OCPU_QPY | / |
| SPI                       | EC600MCNLER06A02M08_OCPU_QPY | / |
| TTS                       | EC600MCNLER06A02M08_OCPU_QPY | / |
| Timer                     | EC600MCNLER06A02M08_OCPU_QPY | / |
| UART                      | EC600MCNLER06A02M08_OCPU_QPY | / |
| USB                       | EC600MCNLER06A02M08_OCPU_QPY | / |
| Wi-Fi Scan                | EC600MCNLER06A02M08_OCPU_QPY | / |
| Encryption and Decryption | EC600MCNLER06A02M08_OCPU_QPY | / |



## About Quectel

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With regional offices and support across the globe, our international leadership is devoted to advancing IoT and building a smarter world.

